Lab 04 – Modbus and OPC UA – See the Difference

Industrial Security (Bachelor) • Maps to Section 04 • 45–60 min

LAB 04

Goal

Read and write Modbus without authentication, then compare with an OPC UA connection. Conclude in plain language why one is fit for the open Internet and one is not.

Part A – Modbus

Step A1 – Honest read

```
docker compose exec student python3 /labs/scripts/modbus_read.py
```

Step A2 – Capture and decode

```
docker compose exec -d student tcpdump -i eth0 -w /labs/captures/lab04-mb.pcap port 502
sleep 1
docker compose exec student python3 /labs/scripts/modbus_read.py
docker compose exec student pkill -INT tcpdump
```

On the host: `tshark -r bachelor/labs/_stack/captures/lab04-mb.pcap -V -Y modbus | less`. *Observation:* note the function code in the request and the byte count in the response.

Step A3 – Inject without authentication

```
docker compose exec student python3 /labs/scripts/modbus_inject.py
```

Confirm in the OpenPLC *Monitoring* page that the coil was written – without any login.

Part B – OPC UA

Step B1 – Browse the namespace

```
docker compose exec student python3 /labs/scripts/opcua_browse.py
```

Step B2 – Capture the OPC UA handshake

```
docker compose exec -d student tcpdump -i eth0 -w /labs/captures/lab04-ua.pcap port 4840
sleep 1
docker compose exec student python3 /labs/scripts/opcua_browse.py
docker compose exec student pkill -INT tcpdump
```

On the host: `tshark -r bachelor/labs/_stack/captures/lab04-ua.pcap -V -d tcp.port==4840,opcua -Y opcua | less`.

Deliverable

Hand in: both PCAPs, an annotated hex dump of one Modbus read, and a 1-page comparison (Modbus vs. OPC UA) across *authentication, integrity, confidentiality, operational cost.*

Caution

Lab containers only. The Modbus injection works because there is no authentication – never run it against equipment you do not own.